

Derivations of the Formulae Used in “Population-Genetics-Informed Neural Networks for a Single-Population Coalescent Under Selection”

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Abstract

The main paper (`paper.tex`) states six numbered equations without reproducing their standard population-genetic derivations, to keep the paper’s own focus on the loss function comparison. This document derives each of the six from first principles, with citations to the primary literature for every non-trivial step, so that a reader unfamiliar with the underlying coalescent and diffusion theory can follow the chain of reasoning end to end. Table 1 maps each derivation here to its equation number and label in the main paper.

Section	Paper Eq. #	Paper label	Topic
1	1	<code>eq:wf-recursion</code>	Discrete-generation selection recursion
2	2	<code>eq:logistic-gen</code>	Continuous-time (weak-selection) limit
3	3	<code>eq:logistic-coalescent</code>	Coalescent-time rescaling, $\alpha = cs_g$
4	4	<code>eq:xtraj</code>	Backward-time hard-sweep trajectory $x(t; s)$
5	5	<code>eq:rate</code>	Sweep-rescaled coalescent rate $\lambda(t, k)$
6	6	<code>eq:kl-sweep</code>	The PGINN KL loss

Table 1: Map from this document’s sections to the main paper’s equations. The main paper numbers equations sequentially by `equation` environment (1–6 in order of appearance); labels are given here since they are unambiguous regardless of any future renumbering.

1 Discrete-generation selection recursion (`eq:wf-recursion`)

Setup

Consider a population reproducing in discrete, non-overlapping generations, with two alleles at a single locus: a favored allele A and a wild-type allele a , at frequencies p and $1 - p$ respectively before selection acts. Under *genic* (additive, no-dominance) selection, each copy of A contributes offspring in proportion to a relative fitness $w_A = 1 + s_g$, and each copy of a in proportion to $w_a = 1$, where $s_g \in [0, 1]$ is the (per-generation) selection coefficient: $s_g = 0$ is neutral, $s_g = 1$ means a leaves no descendants at all. This is the standard fitness parameterization used throughout classical population genetics [Haldane, 1924, Fisher, 1930].

Derivation

In an effectively infinite population (so we can ignore genetic drift and track frequencies deterministically), the frequency of A after selection is the fitness-weighted share of the population:

$$p' = \frac{p \cdot w_A}{p \cdot w_A + (1 - p) \cdot w_a} = \frac{p(1 + s_g)}{p(1 + s_g) + (1 - p)}. \quad (1)$$

Expanding the denominator,

$$p(1 + s_g) + (1 - p) = p + p s_g + 1 - p = 1 + s_g p,$$

so

$$p' = \frac{p(1 + s_g)}{1 + s_g p}, \quad (2)$$

which is exactly Eq. 1 of the main paper. This is the classical discrete-generation map for one-locus genic selection, originally derived by Haldane [1924] in the first of his ten-part series on the mathematical theory of selection, and reproduced in essentially every subsequent population-genetics textbook treatment of directional selection (see also Fisher, 1930 for the contemporaneous treatment via reproductive value).

2 Continuous-time (weak-selection) limit (eq:logistic-gen)

Setup

Eq. 1 is an exact discrete-generation recursion. Many downstream results (diffusion approximations, ODE solutions) are more tractable in continuous time. The standard route from a discrete weak-selection recursion to a continuous-time approximation is the diffusion/mean-field limit formalized by Kimura [1964]: for s_g small, one generation is treated as an infinitesimal time step $d\tau = 1$, and the exact difference equation is replaced by the differential equation it approximates to leading order in s_g .

Derivation

Write the exact change in frequency per generation, $\Delta p \equiv p' - p$, using Eq. 1:

$$\Delta p = \frac{p(1 + s_g)}{1 + s_g p} - p = \frac{p(1 + s_g) - p(1 + s_g p)}{1 + s_g p} = \frac{p s_g (1 - p)}{1 + s_g p}.$$

For weak selection, $s_g p \ll 1$ (true whenever $s_g \ll 1$, since $p \in [0, 1]$), so the denominator is $1 + O(s_g)$ and

$$\Delta p = s_g p(1 - p) + O(s_g^2). \quad (3)$$

Identifying Δp over one generation with the derivative of p with respect to generation-counted time τ (the standard first-order/mean-field diffusion approximation of Kimura, 1964; see also Haldane, 1924 for the earliest use of this continuous approximation in a selection context) gives

$$\frac{dp}{d\tau} = s_g p(1 - p), \quad \tau = \text{time in generations}, \quad (4)$$

which is Eq. 2 of the main paper: the classical logistic equation for allele-frequency change under weak directional selection.

3 Coalescent-time rescaling: $\alpha = c s_g$ (eq:logistic-coalescent)

Setup

Coalescent theory conventionally measures time not in generations but in units of a population-size constant c (generations per coalescent time unit), so that other population-scaled parameters (mutation rate, recombination rate, and here selection) become $O(1)$ quantities as c grows large. This is the same rescaling that turns a per-generation mutation rate μ into the population-scaled parameter $\theta = c\mu$ [Watterson, 1975, Kingman, 1982], and the same rescaling that turns a per-generation selection coefficient into the population-scaled selection intensity familiar from fixation-probability theory [Kimura, 1962] and from the hitchhiking literature [Barton, 1998].

Derivation

Define coalescent time $t = \tau/c$ (so $\tau = ct$). By the chain rule,

$$\frac{dp}{dt} = \frac{dp}{d\tau} \cdot \frac{d\tau}{dt} = \frac{dp}{d\tau} \cdot c.$$

Substituting Eq. 2 ($dp/d\tau = s_g p(1-p)$),

$$\frac{dp}{dt} = c s_g p(1-p) \equiv \alpha p(1-p), \quad \alpha \equiv c s_g, \quad (5)$$

which is Eq. 3 of the main paper. The quantity $\alpha = c s_g$ is exactly the population-scaled selection coefficient conventionally written $2Ns$ or Ns (up to a ploidy/normalization constant folded into c) in the classical fixation-probability and hitchhiking literature [Kimura, 1962, Barton, 1998, Kaplan et al., 1989]: the same non-identifiability argument that makes $\theta = c\mu$, not c and μ separately, the identifiable quantity from a neutral genealogy [Watterson, 1975] applies equally to $\alpha = c s_g$ under selection.

4 Backward-time hard-sweep trajectory $x(t; s)$ (eq:xtraj)

Setup

Eq. 3 (Section 3) is a first-order autonomous ODE and is solvable in closed form (the classical logistic/Verhulst equation). The main paper needs the solution expressed *backward* in time from the present, in terms of the *observed* present-day frequency x_0 rather than the (unobservable) origin time and initial frequency of the sweep, which is the standard reparameterization used in structured- coalescent hitchhiking models [Kaplan et al., 1989, Barton, 1998].

Derivation

Forward solution. Solve $dp/d\tau' = \alpha p(1-p)$ (writing τ' for forward time from the sweep's origin, to distinguish it from the coalescent time τ of Section 3) with initial condition $p(0) = p_0$ by separation of variables:

$$\int \frac{dp}{p(1-p)} = \int \alpha d\tau' \implies \ln \frac{p}{1-p} = \alpha \tau' + \ln \frac{p_0}{1-p_0},$$

using the standard partial-fraction identity $1/[p(1-p)] = 1/p + 1/(1-p)$. Exponentiating and solving for p gives the logistic curve

$$p(\tau') = \frac{p_0 e^{\alpha \tau'}}{1 - p_0 + p_0 e^{\alpha \tau'}}. \quad (6)$$

Backward-time reparameterization. Let T be the (unknown) forward time at which the sweep reaches the present day, so $p(T) = x_0$ is the observed present-day frequency, and define backward time from the present, $t = T - \tau'$ (so $t = 0$ is today, t grows into the past). From Eq. 6 at $\tau' = T$,

$$x_0 = \frac{p_0 e^{\alpha T}}{1 - p_0 + p_0 e^{\alpha T}} \implies p_0 e^{\alpha T} = \frac{x_0(1 - p_0)}{1 - x_0}$$

(solving the defining equation for x_0 for the product $p_0 e^{\alpha T}$). Now write $\tau' = T - t$ in Eq. 6:

$$p_0 e^{\alpha(T-t)} = p_0 e^{\alpha T} e^{-\alpha t} = \frac{x_0(1 - p_0)}{1 - x_0} e^{-\alpha t},$$

and substitute back:

$$x(t) = \frac{\frac{x_0(1 - p_0)}{1 - x_0} e^{-\alpha t}}{(1 - p_0) + \frac{x_0(1 - p_0)}{1 - x_0} e^{-\alpha t}}.$$

The factor $(1 - p_0)$ appears in every term and cancels exactly (for any $p_0 \in (0, 1)$, not only in a limit):

$$x(t) = \frac{\frac{x_0}{1 - x_0} e^{-\alpha t}}{1 + \frac{x_0}{1 - x_0} e^{-\alpha t}} = \frac{x_0 e^{-\alpha t}}{(1 - x_0) + x_0 e^{-\alpha t}}.$$

Multiplying numerator and denominator by $e^{\alpha t}$,

$$x(t; s) = \frac{x_0}{x_0 + (1 - x_0) e^{\alpha t}} = \frac{x_0}{x_0 + (1 - x_0) e^{st}}, \quad (7)$$

using $s \equiv \alpha$: exactly Eq. 5 of the main paper, with $x_0 = 0.999$ fixed as the assumed present-day frequency of a completed (or near-completed) sweep.

On the “ $p_0 \rightarrow 0$ ” remark in the main paper. As the algebra above shows, once $x(t)$ is expressed in terms of the *observed* x_0 , the initial frequency p_0 and origin time T cancel completely and do not appear in the final formula for any $p_0 \in (0, 1)$: the curve shape $x(t; s)$ is determined by (x_0, s) alone. The role of $p_0 \rightarrow 0$ (a single founding copy, $p_0 = 1/c$) is therefore not algebraic but *interpretive*: it is the condition under which this particular family of trajectories (which decays without bound toward $p \rightarrow 0$ as $t \rightarrow \infty$, rather than leveling off at some positive standing frequency) is the physically correct description of the sweep’s full history. This is precisely the hard-sweep-vs.-soft-sweep distinction discussed in the main paper (Section 2.3, `sec:hard-vs-soft` there): a soft sweep from standing variation would instead level off at some $x_{sv} \gg 1/c$ once traced backward far enough [Hermisson and Pennings, 2005], rather than continuing to shrink toward the simulator’s numerical floor $x_{\min}$.

5 Sweep-rescaled coalescent rate $\lambda(t, k)$ (`eq:rate`)

Setup

This equation combines (i) the standard neutral-coalescent pairwise coalescence rate and (ii) the standard device for handling a time-varying effective population size in the coalescent, applied here with the sweep trajectory $x(t; s)$ of Section 4 playing the role of a (backward-time-varying) relative population size.

Derivation

Constant-rate case. In a diploid Wright–Fisher population of effective size N , two specific lineages coalesce (share a parent) in any given generation with probability $\approx 1/(2N)$. With k lineages present, there are $\binom{k}{2} = k(k-1)/2$ distinct pairs, so the total per-generation coalescence rate is $\approx k(k-1)/(4N)$ [Kingman, 1982]. Now measure time in mutation-scaled units $\tau = \mu t_{\text{gen}}$ (i.e. rescale so that a single lineage accumulates mutations at rate 1 per unit τ , the natural units for the population-scaled mutation parameter $\theta = 4N\mu$ [Watterson, 1975]): the per-generation rate $k(k-1)/(4N)$ becomes, per unit τ ,

$$\frac{k(k-1)}{4N} / \mu = \frac{k(k-1)}{4N\mu} = \frac{k(k-1)}{\theta}.$$

This is exactly the $\lambda(t, k) = k(k-1)/\theta_{\text{eff}}$ convention used throughout this project (matching the same $\theta = c\mu$ scaling used in Section 3, with c playing the role of $4N$ or its haploid/diploid analogue depending on convention): the constant c is absorbed into θ , not tracked separately, exactly as Watterson [1975] and Kingman [1982] already require for θ itself to be the identifiable quantity.

Time-varying case. Under the hard-sweep model, the pool of lineages available to coalesce through at backward time t is restricted to the (shrinking, as t grows) fraction $x(t; s)$ of the population that carries the swept allele [Kaplan et al., 1989, Braverman et al., 1995]: effectively, a time-varying relative population size. The standard way to incorporate a time-varying effective population size into the coalescent is to replace the constant rate by an inhomogeneous-Poisson-process rate evaluated at each instant using the population size at that instant [Griffiths and Tavaré, 1994]; applied to the structured hitchhiking coalescent specifically, this gives the sweep-rescaled effective population parameter

$$\theta_{\text{eff}}(t; \theta, s) = \theta x(t; s), \tag{8}$$

and, substituting $\theta_{\text{eff}}(t; \theta, s)$ for θ in the constant-rate formula derived above,

$$\lambda(t, k) = \frac{k(k-1)}{\theta_{\text{eff}}(t; \theta, s)}, \tag{9}$$

which together are Eq. 6 (`eq:rate`) of the main paper. This is the same rescaling used in the classical structured-coalescent approximation of a completed hitchhiking sweep [Kaplan et al., 1989, Barton, 1998, Braverman et al., 1995]: as t grows and $x(t; s)$ shrinks toward the sweep’s origin, θ_{eff} shrinks and λ correspondingly rises, compressing coalescence into a near-star burst near the sweep’s origin.

6 The PGINN loss (`eq:kl-sweep`)

Setup

The final equation scores a network’s predicted $(\hat{\theta}, \hat{s})$ against the true (θ, s) by comparing the coalescent rate each implies, at each of the $K = n - 1$ observed internal-node ages of a sampled genealogy, using the Kullback–Leibler (KL) divergence [Kullback and Leibler, 1951].

Derivation

KL divergence between two exponential distributions. Under the (locally) constant-rate coalescent of Section 5, the waiting time for the next coalescence event among k lineages is exponentially distributed. For two exponential distributions with rates a (density $f_a(x) = a e^{-ax}$) and

b (density $f_b(x) = b e^{-bx}$), the KL divergence of f_b from f_a is, by definition [Kullback and Leibler, 1951],

$$\text{KL}(f_a \parallel f_b) = \int_0^\infty f_a(x) \log \frac{f_a(x)}{f_b(x)} dx.$$

Since $\log[f_a(x)/f_b(x)] = \log(a/b) - (a-b)x$ and $\mathbb{E}_{f_a}[X] = 1/a$,

$$\text{KL}(f_a \parallel f_b) = \log \frac{a}{b} - (a-b) \cdot \frac{1}{a} = \log \frac{a}{b} + \frac{b}{a} - 1.$$

This is a standard closed-form result for the exponential family (consistent with the general KL definition of Kullback and Leibler, 1951; any information-theory text derives the same identity for the exponential case by the integral above). Setting $a = \hat{\lambda}_k$ (the network’s predicted rate) and $b = \lambda_k$ (the rate implied by the true parameters) gives the per-interval term

$$\text{KL}(f_{\hat{\lambda}_k} \parallel f_{\lambda_k}) = \log \frac{\hat{\lambda}_k}{\lambda_k} + \frac{\lambda_k}{\hat{\lambda}_k} - 1,$$

i.e. $\text{KL}(\text{predicted} \parallel \text{true})$: the network’s predicted exponential distribution is the first (“reference”) argument, so the loss penalizes predictions whose implied waiting-time distribution diverges from the true one, evaluated under the *predicted* law – the direction that keeps the loss finite and well-behaved even for a very confident but wrong prediction, since the expectation in the definition of $\text{KL}(f_{\hat{\lambda}_k} \parallel f_{\lambda_k})$ is taken under $\hat{\lambda}_k$, the quantity the network controls.

From general rates to the sweep model. Substituting the sweep-rescaled rate of Section 5,

$$\hat{\lambda}_k = \frac{k(k-1)}{\hat{\theta} x(t_k; \hat{s})}, \quad \lambda_k = \frac{k(k-1)}{\theta_{\text{true}} x(t_k; s_{\text{true}})},$$

evaluated at the k -th observed internal-node age t_k of the sample’s own genealogy (sorted descending, $k = 2, \dots, K+1$ lineages present just before that coalescence), and averaging over the K intervals, gives

$$\mathcal{L}_{\text{PGINN}} = \frac{1}{K} \sum_{k=2}^{K+1} \left[\log \frac{\hat{\lambda}_k}{\lambda_k} + \frac{\lambda_k}{\hat{\lambda}_k} - 1 \right], \quad (10)$$

which is Eq. 7 (eq:kl-sweep) of the main paper.

An explicit approximation, flagged for what it is. The closed-form KL divergence above is an exact property of *two fixed-rate* exponential distributions. Here, however, $\hat{\lambda}_k$ and λ_k are themselves time-varying (through $x(t; \cdot)$, Section 4) and are evaluated only at the single observed age t_k rather than integrated over the interval as a true inhomogeneous-Poisson-process likelihood would require [Griffiths and Tavaré, 1994, Kaplan et al., 1989]. Eq. 7 is therefore a *locally frozen-rate* approximation: at each interval it treats the rate as if it were constant at its value at the observed t_k , and scores the resulting (correctly-derived) exponential-vs.-exponential KL divergence at that frozen rate. This is exactly the caveat the main paper’s Discussion states explicitly (“The KL term also approximates the true coalescent-with-selection likelihood by evaluating the sweep-rescaled exponential rate at the observed ages rather than integrating the full inhomogeneous-Poisson-process likelihood over each interval”); this document simply makes precise which approximation step that sentence refers to.

Reduction to the neutral case. At $s = 0$ (and $\hat{s} = 0$), $x(t; 0) = x_0$ for all t (Eq. 5, Section 4), a constant, so $\hat{\lambda}_k = k(k - 1)/(\hat{\theta} x_0)$ and $\lambda_k = k(k - 1)/(\theta_{\text{true}} x_0)$: the x_0 factor is common to both and Eq. 7 reduces to the ordinary single-population neutral coalescent KL loss in θ alone, up to the fixed rescaling constant $x_0 = 0.999$ – confirming that the selection-augmented loss nests the neutral loss as a special case, as the main paper states in Section 2.4 (`sec:loss`) there.

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